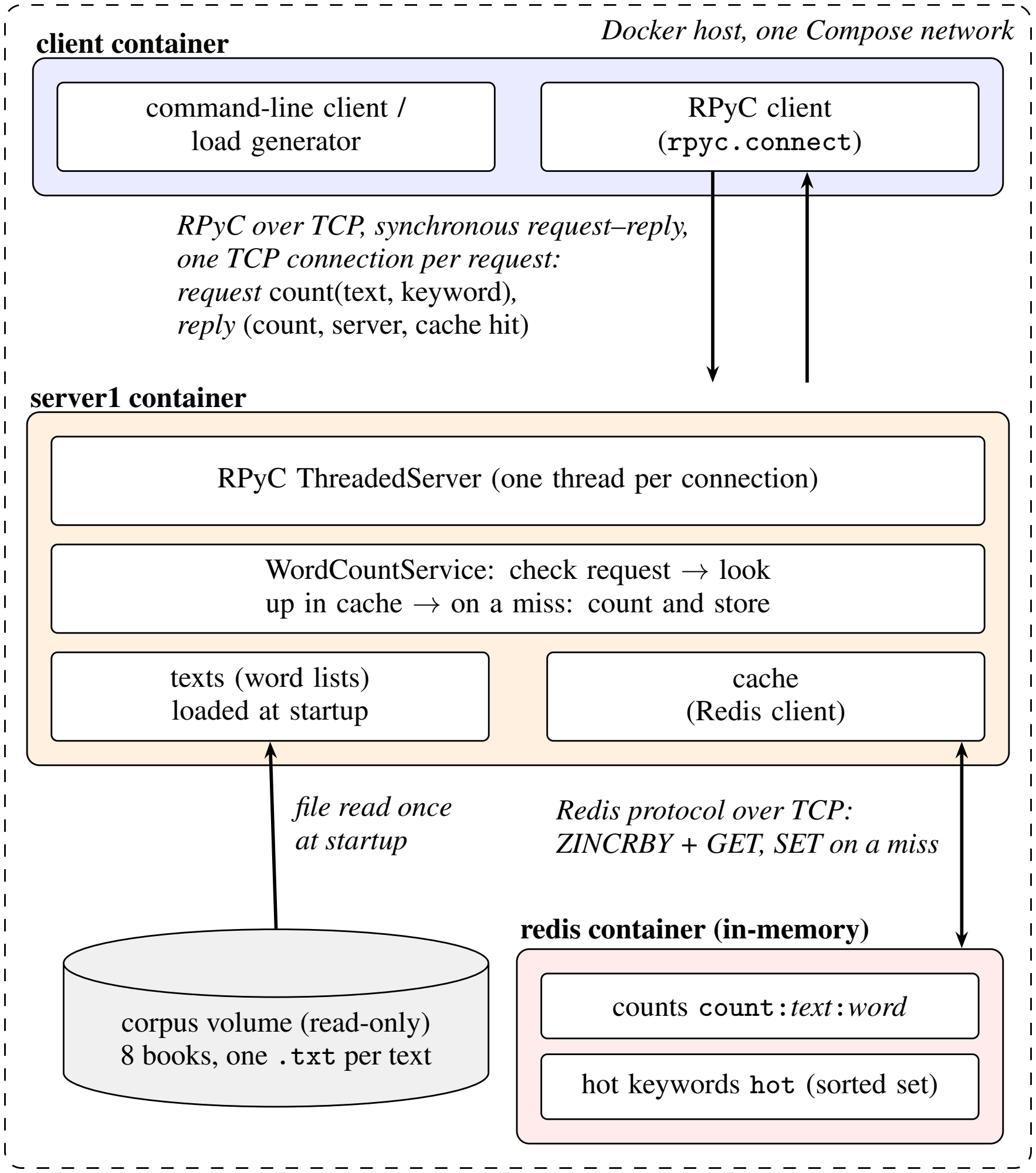
Docker host, one Compose network
client container
command-line client / load generator
RPyC client (rpyc.connect)
RPyC over TCP, synchronous request–reply, one TCP connection per request:
request count(text, keyword),
reply (count, server, cache hit)
server1 container
RPyC ThreadedServer (one thread per connection)
WordCountService: check request → look up in cache → on a miss: count and store
texts (word lists) loaded at startup
cache (Redis client)
file read once at startup
Redis protocol over TCP: ZINCRBY + GET, SET on a miss
redis container (in-memory)
corpus volume (read-only) 8 books, one .txt per text
counts count:text:word
hot keywords hot (sorted set)